Cyclohexane Reactor Ver 1 · operator training simulator

Safety requirements specification of the SIFs

Document no. CHX-V1-IN-SRS-001 · **Discipline** Instrumentation and control

Client Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University (teaching)

• Safety requirements specification (IEC 61511-1:2016 clause 10) for the 8 functions of SIS-100: the six trips, the emergency shutdown HS-100 and the hydrogen start permissive.
• As found: **3 OK · 0 Check · 5 Gap**. SIL targets from the LOPA scenarios; PFDavg from the LOPA block diagrams (proof test T = 1 yr, β = 0); process safety times from 12 demand cases run on the simulator model, each twice (trips disabled, then the SIS acting).
• Set points and delays are those of the model (K, TRIPDLY), identical to the C&E matrix CHX-V1-IN-CE-001. Response = sensor 0.5 + logic 0.25 + trip delay + final element 2 s (assumed).
• Same data as the SRS of SIF tab of the simulator: tabs/cem/sf.json (requirements text, common requirements), tabs/srs/sr_pst.json (model runs), LOPA tab (PFD, SIL).

Revisions

Rev	Date	Description	Originator	Checker	Approver
0	2026-10-26	Issued for teaching	Apinan Soottitantawat		

This document is generated for teaching from the simulator model of the unit. It is not issued for construction; every number comes from the model or is labelled assumed / typical practice. No client, contractor or vendor is implied.

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1 · Scope and position in the safety life cycle

This specification states, for every safety instrumented function (SIF) of Cyclohexane Reactor Ver 1, the hazard it protects against, the demands, the safe state, the sensor and its setting, the time allowed, the final elements, the required safety integrity, and how the function is armed, bypassed, reset and proof-tested. It is the document the instrument engineer designs the SIS-100 logic and hardware from, the C&E matrix is checked against, and the maintenance team tests to.
IEC 61511-1 orders the work as a life cycle (figure 1). The hazard and risk assessment (clause 8: HAZID, HAZOP, LOPA) finds what can go wrong and how much risk reduction is missing; the allocation (clause 9: safety layers, LOPA) decides which layer provides it. **Clause 10, the SRS, is the hinge**: everything before it is analysis; design, the logic program, installation, validation, operation and proof testing (clauses 11–16) are built and checked against it, and every modification (clause 17) goes back to the risk assessment.

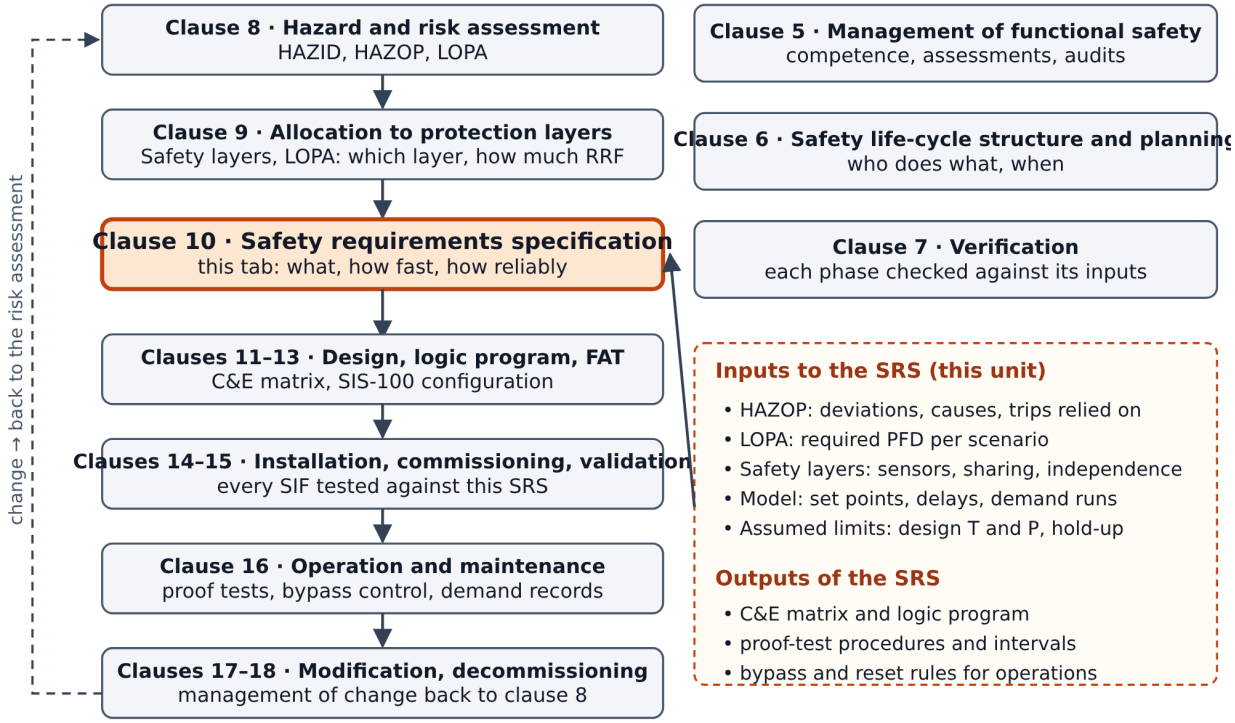


Figure 1 — IEC 61511 safety life cycle; this document is clause 10 (drawn by the SRS tab).

1.1 Contents required by clause 10.3 and where they are

Clause 10.3 item	Where in this document
Description of each SIF, hazard, safe state, demand sources and demand rate	§3 per SIF: Hazard, Demand sources, Safe state
Process measurements, trip points, action on each final element, manual shutdown	§2 SIF list; §3 Sensor, Set point, Final elements; SIF-7 HS-100
Mode of operation (low demand), SIL and PFD target, how they were derived	§2; §3 SIL target and basis (LOPA scenario, target / frequency)

Clause 10.3 item	Where in this document
Response time and process safety time	§3 Process safety time (model runs, timeline figures); §4 Response-time budget
Proof-test interval, reset, bypass and start-up arming	§3 Arming, Reset, Proof test; §4 Bypass, Reset, Proof test
Spurious-trip rate, behaviour on detected faults, de-energise to trip, HFT	§4 Common requirements
Failure data and architecture used for the PFD	§3 Reliability block diagram and subsystem table
Verification against the requirements, gaps and recommendations	§2 verdict columns; §5 Findings

1.2 Document information and references

Item	Value
Client	Department of Chemical Engineering, Chulalongkorn University
Project	Cyclohexane Reactor Ver 1 · operator training simulator (benzene hydrogenation, 300 kt/y)
Document	CHX-V1-IN-SRS-001 · Safety requirements specification of the SIFs · Rev 0 Issued for teaching
CHX-V1-PID-001	Screen P&ID P&ID-001 (P&ID tab of the simulator)
CHX-V1-PFD-001	Process flow diagram
CHX-V1-IN-SRS-001	Safety requirements specification of the SIFs (this document)
CHX-V1-SA-HZP-001	HAZOP report
CHX-V1-IN-CE-001	Cause and effect matrix of SIS-100, alarms and permissives (same set points, delays and effects)
CHX-V1-PE-RLB-001	Relief load basis (PSHH-101 33.5 bar g below the 36 bar g relief set; relief cases with no SIS credit)
CHX-V1-IN-INL-001	Instrument list (trip transmitters, XV-101, XV-105, analyser AI-100)
CHX-V1-PM-DOC-001	Document register and engineering queries

1.3 Basis and assumptions

Topic	Basis
Set points and delays	Model constants K.T1hh 220 °C, K.T3hh 220 °C, K.P1hh 33.5 bar g, K.L4hh 90 %; TRIPDLY 5, 5, 3, 10, 10, 5 s (the model acts on the next 0.25 s step).
SIL target	In each LOPA scenario that credits the SIF, the SIF (and the layers depending on its sensor) is switched off; the causes it is credited for give f (/yr); PFD required = target / f. SIL = band of that number, at least SIL 1 for every SIS function; the requirement is the stricter of the band limit and the LOPA number. Functions outside the LOPA: SIL 1 assumed.
PFDavg	Σ subsystem PFD; 1oo1 λT/2, 1oo2 $(\lambda T)^2/3$, 2oo3 $(\lambda T)^2$, NooN Σ λT/2; λDU from the LOPA tab (SIS-grade rates); β = 0, proof-test coverage 100 % (course method).
Process safety time	Two runs per demand case on a copy of the design state: unprotected (every trip set out of reach and the trip routine stubbed) and protected (SIS as designed). PST = t(hazard) − t(trip point first crossed).
Hazard limits (assumed)	R-101 250 °C (design, assumed) or level below 25 %; R-103 bed 260 °C; loop 36 bar g (relief set); V-104 100 % with K-101 running; benzene hold-up 11.8 kmol (LOPA tab).
Response time (assumed)	Sensor 0.5 s + logic 0.25 s + trip delay + final element 2 s. Verdict: OK response ≤ ½ PST; Check ≤ PST; Gap longer, or the hazard before the trip point, or a demand the SIF cannot see, or the hazard still reached with the SIS acting.

2 · SIF list

8 functions in SIS-100 (logic solver separate from the DCS). The verdict columns judge the PFDavg against the requirement (**PFD**), the response against the process safety time for the demands the function sees (**Time**), and whether it sees every demand of the LOPA and HAZOP (**Coverage**); the overall verdict is the worst of the three. As found: 3 OK, 0 Check, 5 Gap.

SIF	Tag	Function	Set point	Delay s	SIL target · PFD req.	SIL basis	PFDavg	RRF	PST	Resp.	PFD	Time	Cover.	Verdict
SIF-1	TSHH-100	R-101 temperature high-high → unit trip	> 220 °C	5	SIL 1 · ≤ 0.1	LOPA L1: none needed; SIS minimum	5.58×10^{-3}	179	3 min 46 s	7.75 s	OK	OK	Gap	Gap
SIF-2	TSHH-107	R-103 bed temperature high-high → unit trip	> 220 °C (hottest of 5)	5	SIL 1 · ≤ 0.1	LOPA L5: none needed; SIS minimum	5.58×10^{-3}	179	35 min 57 s	7.75 s	OK	OK	OK	OK
SIF-3	PSHH-101	R-101 pressure high-high → make-up isolated, unit trip	> 33.5 bar g	3	SIL 2 · ≤ 6.67×10^{-3}	LOPA L3: stricter than the band	0.0175	57	16.25 s	5.75 s	Gap	OK	OK	Gap
SIF-4	XS-101	Agitator M-101 stopped with feed → unit trip	motor stopped with XV-101 open	10	SIL 1 · ≤ 0.02	LOPA L2: stricter than the band	0.0106	95	1 min 31 s	12.75 s	OK	OK	Gap	Gap
SIF-5	XS-102	K-101 stopped with feed → benzene isolated	motor stopped with XV-101 open	10	SIL 1 · ≤ 0.02	LOPA L2: stricter than the band	0.0225	44	1 min 03 s	12.75 s	Gap	OK	Gap	Gap
SIF-6	LSHH-104	V-104 level high-high → K-101 stops	> 90 %	5	SIL 1 · ≤ 0.1	not in LOPA (assumed)	0.0105	95	3 min 34 s	7.75 s	OK	OK	OK	OK
SIF-7	HS-100	ESD pushbutton → unit trip and depressuring	pressed	0	SIL 1 · ≤ 0.1	not in LOPA (assumed)	0.037	27	n/a	n/a	OK	n/a	n/a	OK
PERM-H2	AI-100	Start permissive: XV-105 opens only with loop O2 below the limit	≥ 1 % blocks XV-105	—	SIL 2 · ≤ 6.67×10^{-3}	LOPA L4: stricter than the band	0.0313	32	n/a	n/a	Gap	n/a	n/a	Gap

Set points and delays from the model (K, TRIPDLY); PFDavg from the LOPA block diagrams; SIL target from the LOPA scenarios with the function switched off; PST (governing demand = the shortest PST among the demands the function sees) and response from the model runs of §3. SIF-7 and PERM-H2 have no process safety time (manual function; a permissive acts before the hazardous state exists).

3 · Requirements per SIF

One sub-section per function: the requirement table (IEC 61511-1 10.3), the reliability block diagram with λDU and PFD per subsystem, and the process safety time measured on the model for every demand case, with the timeline of each case (numbered events on a square-root time axis; for cases with a PST, the response bar against PST and ½ PST). Figures are the drawings of the SRS tab.

3.1 SIF-1 · TSHH-100 — R-101 temperature high-high → unit trip

Requirement	Specification
Hazard (HAZOP N2.1, N3, N4.1, LOPA L1)	R-101 runaway: the slurry heats toward its boiling point (about 258 °C at reactor pressure) and above the R-101 design temperature (250 °C, assumed); hot hydrogen and cyclohexane vapour overload the overhead, a flange or seal fails, fire.
Demand sources	• Pump-around flow lost (running pump stops, spare not started; FV-104 path blocked) — LOPA 0.1 /yr • TIC-101 / FIC-104 control loop fails — 0.1 /yr • E-101 water side lost (D-101 drum level low) • Feed raised too fast after light-off — 0.02 /yr Low-demand mode (frequencies from the LOPA tab).
Safe state	Reaction heat stopped: benzene and make-up hydrogen isolated, K-101 stopped; E-101 on full cooling and the R-102 jacket on cooling as BPCS back-up actions.
Sensor and voting	TE-100 with its own SIL-rated transmitter (TI-100), separate from the TIC-101 element TE-101; 1oo1
Set point	> 220 °C · pre-alarm TI-100 HI 210, TIC-101 HI 208 (TI-100 on the trip element TE-100; TIC-101 on TE-101)
Delay	5 s (the model acts on the next 0.25 s step)
SIL target and basis	SIL 1, PFD ≤ 0.1. LOPA L1 (severity 4, target 10^{-3} /yr). Causes credited to this function: Pump-around flow lost (running pump stops, standby not started); TIC-101 / FIC-104 loop fails (TE-101 reads low, FV-104 stuck). Without it they occur at 0.01 /yr, so PFD ≤ 10^{-3} / 0.01 = 0.1 (no SIL needed by the LOPA; SIL 1 is the minimum for a function in the SIS). Other causes in L1 that this function is not credited for: Cooling water lost to the reflux condenser E-103 (does not see the boil-down: TI-100 stays near 200 °C until the level is gone (model)).
PFDavg (computed)	5.58×10^{-3} (RRF 179, SIL 2 by PFD) OK · proof-test interval T = 1 yr, β = 0
Final elements	• XV-101 benzene SIS valve closes (fail closed, SOV) • XV-105 make-up hydrogen SIS valve closes (fail closed, SOV) • K-101 stopped (MCC stop relay) • BPCS actions, not credited: FIC-101, FIC-103, PIC-105 to MAN 0 %; E-101 to COOL with FV-104 forced 100 %; R-102 jacket to COOL
Arming and bypass	Always armed (also during start-up heating with MP steam). Bypass: common requirements, §4.
Reset	Latched in SIS-100; manual reset on the SIS panel, refused while a non-motor trip condition is still present or TI-100 is above 210 °C; after the reset hydrogen and benzene stay isolated until they are admitted again through their permissives.
Proof test	Every year: inject at the element (calibrator bath or simulated signal at the transmitter), trip at 220 °C after 5 s; confirm XV-101 and XV-105 close within 2 s and K-101 stops; check the first-out.
Process safety time and response	Governing demand: Pump-around lost and the reflux condenser E-103 without cooling water (HV-113 shut) at the same time. PST 3 min 46 s (trip point 1 min 30 s → hazard 5 min 16 s); response 7.75 s ≤ ½ PST. Hazard criterion: R-101 above 250 °C (design temperature, assumed) or R-101 level below 25 % (impeller and sparger uncovered, assumed). Time verdict OK.
Finding	Sees every loss of heat removal on the pump-around side, but not R-101 boiling down when the reflux condenser E-103 loses its cooling water: the temperature stays near 200 °C while the level falls (LOPA L1 cause 2). Recommendation R05 (LSLL-101 on its own element) or R06 (E-103 low cooling-water flow trip). Overall verdict Gap.

Reliability block diagram (LOPA tab, key tshh100) — Either final action ends the heat release: no benzene, or no hydrogen.

Subsystem	Architecture	λDU /yr	PFD	Share of PFDavg
TE/TT-100 (own element)	1oo1	0.01	5×10^{-3}	90 %
SIS-100 logic solver	1oo1	10^{-3}	5×10^{-4}	9 %
XV-101 + SOV or K-101 stop	1oo2	0.024 / 0.01	8×10^{-5}	1 %
SIF			5.58×10^{-3}	T = 1 yr

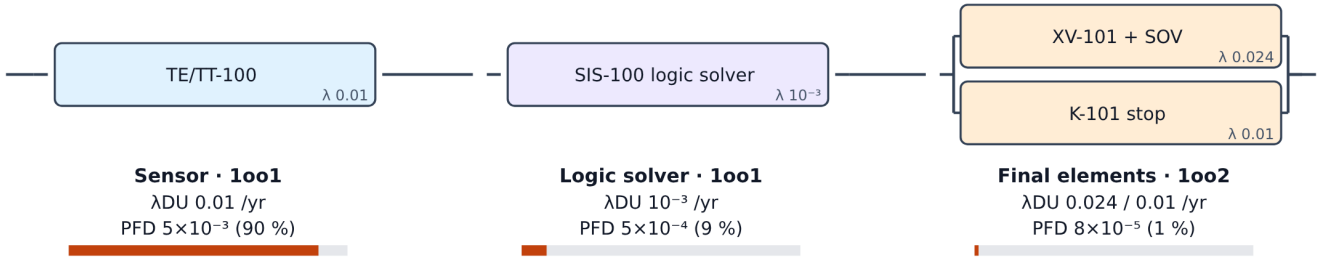
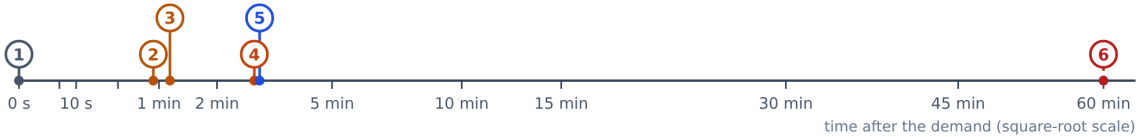


Figure 2 — SIF-1 TSHH-100 reliability block diagram (PFDavg = Σ subsystem PFD).

Process safety time on the model — 3 demand case(s); hazard criterion: R-101 above 250 °C (design temperature, assumed) or R-101 level below 25 % (impeller and sparger uncovered, assumed).

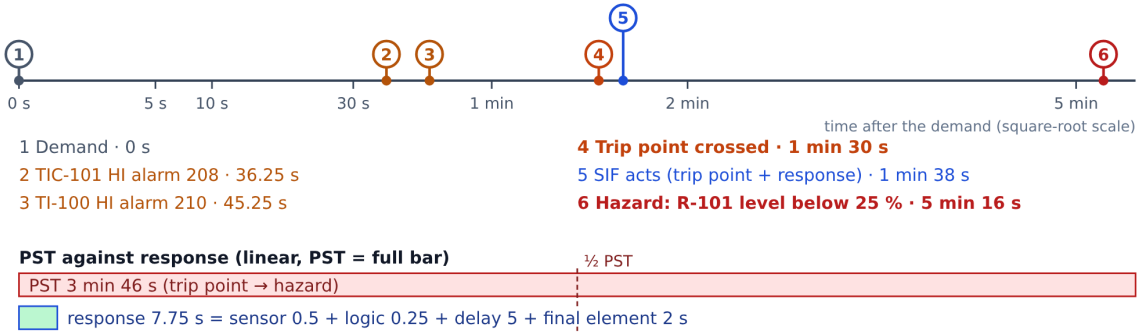
Demand (applied at t = 0)	Trip point crossed	Hazard (no trip)	PST	Response	With the SIS acting	Verdict
Pump-around lost: running pump P-101A stops, spare not started	2 min 50 s	not in 60 min	—	7.75 s	trip 2 min 55 s (TSHH-100); TI-100 max 220.3 °C, level min 50.8 %; hazard not reached	OK hazard not reached in 60 min without any trip (peak 224.032 °C)
Pump-around lost and the reflux condenser E-103 without cooling water (HV-113 shut) at the same time [governing]	1 min 30 s	5 min 16 s R-101 level below 25 %	3 min 46 s	7.75 s	trip 1 min 36 s (TSHH-100); TI-100 max 221.1 °C, level min 48.8 %; hazard not reached	OK response ≤ ½ PST
Reflux condenser E-103 without cooling water (HV-113 shut), pump-around running [not seen]	—	48 min 11 s R-101 level below 25 %	—	7.75 s	trip 79 min 14 s (TSHH-100); TI-100 max 221.8 °C, level min 3.7 %; hazard reached 48 min 11 s	Gap the SIF does not see this demand: its trip point is never crossed before the hazard; the model still reaches the hazard with the SIS acting



1 Demand · 0 s
2 TIC-101 HI alarm 208 · 55.25 s
3 TI-100 HI alarm 210 · 1 min 10 s
4 Trip point crossed · 2 min 50 s
5 SIF acts (trip point + response) · 2 min 57 s
6 Hazard not reached in 60 min · 60 min 00 s

The hazard is not reached without any trip: no process safety time for this demand.

Figure 3 — TSHH-100 timeline: Pump-around lost: running pump P-101A stops, spare not started.



1 Demand · 0 s
2 TIC-101 HI alarm 208 · 36.25 s
3 TI-100 HI alarm 210 · 45.25 s
4 Trip point crossed · 1 min 30 s
5 SIF acts (trip point + response) · 1 min 38 s
6 Hazard: R-101 level below 25 % · 5 min 16 s

Figure 4 — TSHH-100 timeline: Pump-around lost and the reflux condenser E-103 without cooling water (HV-113 shut) at the same time (governing).



1 Demand · 0 s
2 Hazard: R-101 level below 25 % · 48 min 11 s

The trip point is never crossed: this SIF does not see the demand.

Figure 5 — TSHH-100 timeline: Reflux condenser E-103 without cooling water (HV-113 shut), pump-around running (not seen by this SIF).

3.2 SIF-2 · TSHH-107 — R-103 bed temperature high-high → unit trip

Requirement	Specification
Hazard (HAZOP N7.1, N1.2, N5.1, LOPA L5)	Hot spot in the adiabatic R-103 bed (about 9 °C per 1 % benzene reaching it): shell or outlet flange above the design temperature (260 °C, assumed) at 29 bar g, hydrogen release, fire.
Demand sources	• Benzene slip from R-101 / R-102 (loss of mixing, low catalyst activity) — 0.1 /yr • Benzene feed above what R-101 converts (FIC-101 loop) — 0.1 /yr • Reflux condenser without cooling water — 0.1 /yr Low-demand mode (frequencies from the LOPA tab).
Safe state	Benzene and hydrogen to the bed stopped (XV-101, XV-105 closed, K-101 stopped) so the hot spot burns out without new benzene.
Sensor and voting	TE-107A…E, one element per bed cell; the hottest of five is compared with the setting. A hot spot is seen by the one element in it: 1oo1 per zone. Elements shared with the TI-107 indication and alarm.
Set point	> 220 °C (hottest of 5) · pre-alarm TI-107 HI 185 (on the same elements as the trip)
Delay	5 s (the model acts on the next 0.25 s step)
SIL target and basis	SIL 1, PFD ≤ 0.1. LOPA L5 (severity 4, target 10^{-3} /yr). Causes credited to this function: Loss of mixing in R-101 lets benzene slip forward (motor stop or drive failure); Benzene feed above what R-101 converts (FIC-101 loop fails, FI-101 reads low); Reflux condenser E-103 without cooling water. Without it they occur at 6×10^{-3} /yr, so PFD ≤ 10^{-3} / 6×10^{-3} = 0.167 (no SIL needed by the LOPA; SIL 1 is the minimum for a function in the SIS).
PFDavg (computed)	5.58×10^{-3} (RRF 179, SIL 2 by PFD) OK · proof-test interval T = 1 yr, β = 0
Final elements	• XV-101 benzene SIS valve closes (fail closed, SOV) • XV-105 make-up hydrogen SIS valve closes (fail closed, SOV) • K-101 stopped (MCC stop relay) • BPCS actions, not credited: FIC-101, FIC-103, PIC-105 to MAN 0 %; E-101 to COOL with FV-104 forced 100 %; R-102 jacket to COOL
Arming and bypass	Always armed. Bypass: common requirements, §4.
Reset	Latched in SIS-100; manual reset on the SIS panel, refused while a non-motor trip condition is still present or TI-100 is above 210 °C; after the reset hydrogen and benzene stay isolated until they are admitted again through their permissives.
Proof test	Every year: each of the five elements in turn, trip at 220 °C after 5 s; valve stroke and K-101 stop as for SIF-1. Compare the five cells on every shift (at design they agree within a fraction of a degree) to find a drifting element.
Process safety time and response	Governing demand: Benzene slip to R-103: mixing in R-101 lost while the agitator motor keeps running (LOPA L5 cause 1). PST 35 min 57 s (trip point 44 min 34 s → hazard 80 min 31 s); response 7.75 s ≤ ½ PST. Hazard criterion: Hottest R-103 bed temperature above 260 °C (design temperature, assumed). Time verdict OK.
Finding	The trip elements are also the TI-107 indicators and the alarm, so a failed element blinds alarm and trip for its cell together. Recommendation SL-R5: separate SIS elements, 2oo3 per hot zone (TSHH-107S). Overall verdict OK.

Reliability block diagram (LOPA tab, key tshh107) — Five cells, but a hot spot is seen by the one element in it: 1oo1 for that zone.

Subsystem	Architecture	λDU /yr	PFD	Share of PFDavg
TE/TT-107 at the hot spot (shared with TI-107)	1oo1	0.01	5×10^{-3}	90 %
SIS-100 logic solver	1oo1	10^{-3}	5×10^{-4}	9 %
XV-101 + SOV or K-101 stop	1oo2	0.024 / 0.01	8×10^{-5}	1 %
SIF			5.58×10^{-3}	T = 1 yr

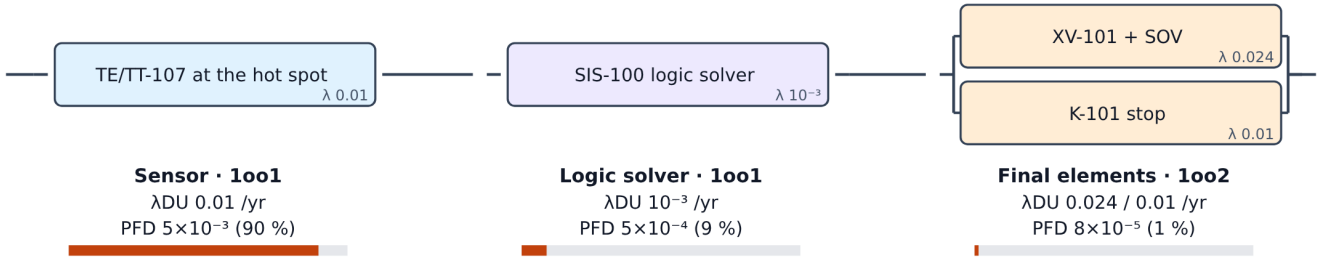


Figure 6 — SIF-2 TSHH-107 reliability block diagram (PFDavg = Σ subsystem PFD).

Process safety time on the model — 2 demand case(s); hazard criterion: Hottest R-103 bed temperature above 260 °C (design temperature, assumed).

Demand (applied at t = 0)	Trip point crossed	Hazard (no trip)	PST	Response	With the SIS acting	Verdict
Benzene slip to R-103: mixing in R-101 lost while the agitator motor keeps running (LOPA L5 cause 1) [governing]	44 min 34 s	80 min 31 s bed above 260 °C	35 min 57 s	7.75 s	trip 44 min 46 s (TSHH-107); max 220.1 °C; hazard not reached	OK response ≤ ½ PST
Benzene feed above what R-101 converts: FV-101 driven fully open (FIC-101 output 100 %)	—	not in 120 min	—	7.75 s	no trip; max 217.8 °C; hazard not reached	OK no demand: the trip point is not reached and neither is the hazard in 120 min

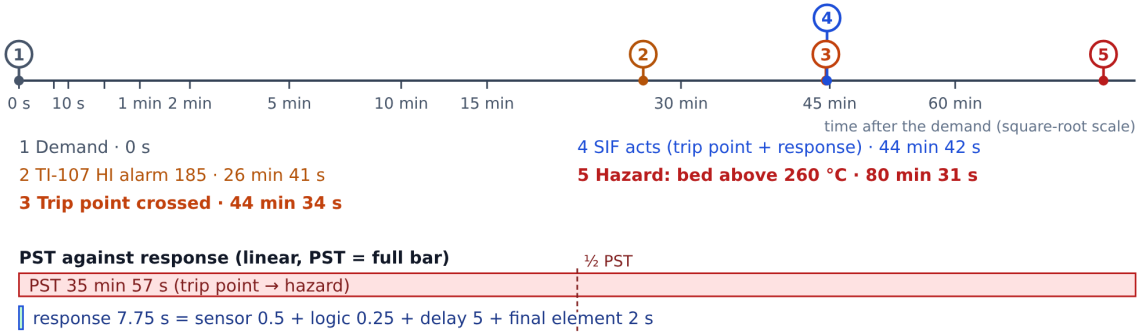


Figure 7 — TSHH-107 timeline: Benzene slip to R-103: mixing in R-101 lost while the agitator motor keeps running (LOPA L5 cause 1) (governing).

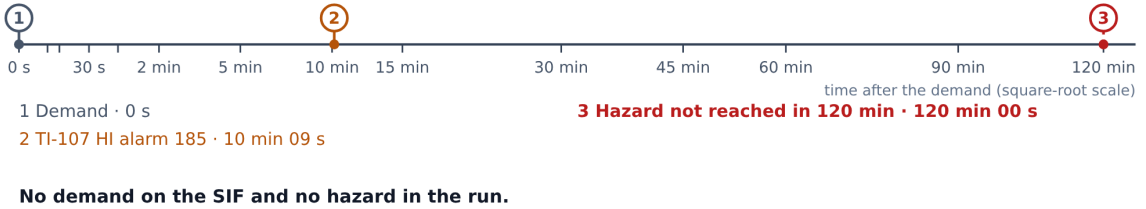


Figure 8 — TSHH-107 timeline: Benzene feed above what R-101 converts: FV-101 driven fully open (FIC-101 output 100 %).

3.3 SIF-3 · PSHH-101 — R-101 pressure high-high → make-up isolated, unit trip

Requirement	Specification
Hazard (HAZOP N2.7, N8.4, LOPA L3)	Hydrogen loop above its design pressure (36 bar g, assumed = relief set): flange or vessel failure on R-101, E-103 / V-103, R-102, R-103 or V-104, hydrogen released at 30 bar g or more, fire or explosion.
Demand sources	• Make-up valve PV-105 driven open (PIC-105 output high or PT-105 reading low) — 0.1 /yr × 0.1 • Make-up supply regulator upstream fails high — 0.1 /yr • Gas path blocked downstream of R-101 with K-101 running (dead-head about 40.5 bar g; not in the model) — 0.01 /yr Low-demand mode (frequencies from the LOPA tab).
Safe state	Make-up hydrogen isolated by XV-105 and K-101 stopped, so the loop pressure stops rising; the benzene feed is stopped with the unit trip.
Sensor and voting	PT-101 at R-101 (own SIL-rated transmitter; PIC-105 controls on PT-105 at V-104); 1oo1
Set point	> 33.5 bar g · pre-alarm PIC-105 HI 30.5 (on PT-105 at V-104, not the trip transmitter)
Delay	3 s (the model acts on the next 0.25 s step)
SIL target and basis	SIL 2, PFD ≤ 6.67×10^{-3}. LOPA L3 (severity 5, target 10^{-4} /yr). Causes credited to this function: Make-up valve PV-105 driven open (PIC-105 output high or PT-105 reading low); Make-up supply regulator upstream fails high; Gas path blocked downstream of R-101 with K-101 running. Without it they occur at 0.015 /yr, so PFD ≤ 10^{-4} / 0.015 = 6.67×10^{-3} → SIL 2, and stricter than the SIL 2 band limit 0.01.
PFDavg (computed)	0.0175 (RRF 57, SIL 1 by PFD) Gap · proof-test interval T = 1 yr, β = 0
Final elements	• XV-101 benzene SIS valve closes (fail closed, SOV) • XV-105 make-up hydrogen SIS valve closes (fail closed, SOV) • K-101 stopped (MCC stop relay) • BPCS actions, not credited: FIC-101, FIC-103, PIC-105 to MAN 0 %; E-101 to COOL with FV-104 forced 100 %; R-102 jacket to COOL
Arming and bypass	Always armed. Bypass: common requirements, §4.
Reset	Latched in SIS-100; manual reset on the SIS panel, refused while a non-motor trip condition is still present or TI-100 is above 210 °C; after the reset hydrogen and benzene stay isolated until they are admitted again through their permissives.
Proof test	Every year: pressure applied at PT-101, trip at 33.5 bar g after 3 s; XV-105 stroke-time and seat-leak test (it is the only make-up isolation); K-101 stop.
Process safety time and response	Governing demand: PV-105 fully open with the make-up supply regulator failed: header at 45 bar g (assumed source pressure). PST 16.25 s (trip point 21.5 s → hazard 37.75 s); response 5.75 s ≤ ½ PST. Hazard criterion: PI-101 above 36 bar g (hydrogen-loop design pressure = PSV set, assumed). Time verdict OK.
Finding	One valve XV-105 sets most of the PFD and the LOPA (L3, severity 5) asks for more than a 1oo1 valve gives. Recommendation LP-R1: a second make-up shut-off valve XV-105B in series (1oo2). Relief valves PSV-101 / PSV-104 (SL-R8, SL-R10, R02) remove the scenario almost entirely; a PI-101 pre-alarm (SL-R1, R21) helps the operator but shares PT-101. Overall verdict Gap.

Reliability block diagram (LOPA tab, key pshh101) — Only XV-105 stops the make-up hydrogen; one valve sets the PFD.

Subsystem	Architecture	λDU /yr	PFD	Share of PFDavg
PT-101 (own transmitter)	1oo1	0.01	5×10^{-3}	29 %
SIS-100 logic solver	1oo1	10^{-3}	5×10^{-4}	3 %
XV-105 + SOV	1oo1	0.024	0.012	69 %
SIF			0.0175	T = 1 yr

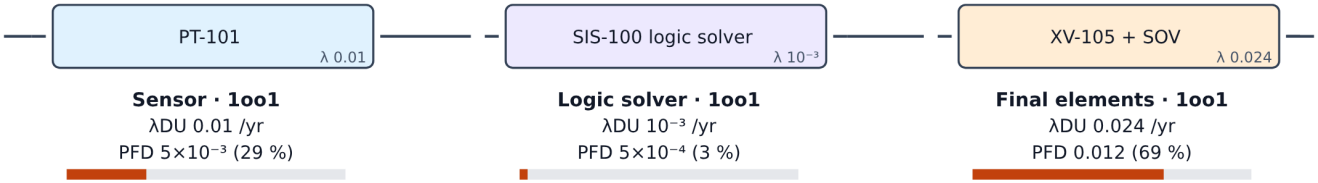


Figure 9 — SIF-3 PSHH-101 reliability block diagram (PFDavg = Σ subsystem PFD).

Process safety time on the model — 2 demand case(s); hazard criterion: PI-101 above 36 bar g (hydrogen-loop design pressure = PSV set, assumed).

Demand (applied at t = 0)	Trip point crossed	Hazard (no trip)	PST	Response	With the SIS acting	Verdict
Make-up valve PV-105 driven fully open (PIC-105 output 100 %), make-up header at its design pressure	53 s	not in 30 min	—	5.75 s	trip 56.25 s (PSHH-101); max 33.58 bar g; hazard not reached	OK hazard not reached in 30 min without any trip (peak 34.498 bar g)
PV-105 fully open with the make-up supply regulator failed: header at 45 bar g (assumed source pressure) [governing]	21.5 s	37.75 s PI-101 above 36 bar g	16.25 s	5.75 s	trip 24.75 s (PSHH-101); max 34.03 bar g; hazard not reached	OK response ≤ ½ PST



The hazard is not reached without any trip: no process safety time for this demand.

Figure 10 — PSHH-101 timeline: Make-up valve PV-105 driven fully open (PIC-105 output 100 %), make-up header at its design pressure.

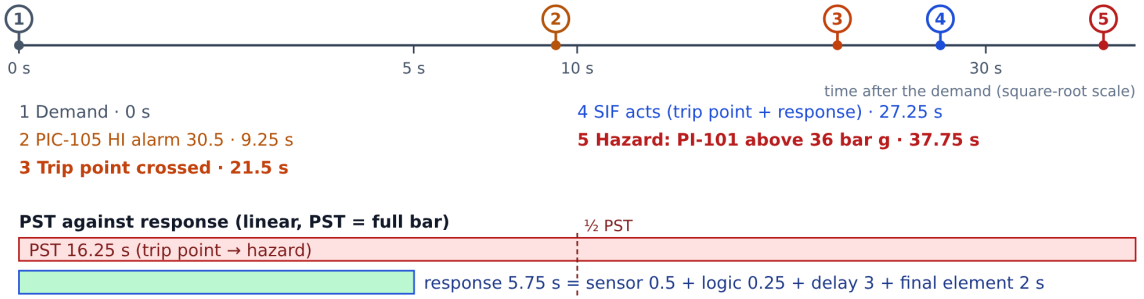


Figure 11 — PSHH-101 timeline: PV-105 fully open with the make-up supply regulator failed: header at 45 bar g (assumed source pressure) (governing).

3.4 SIF-4 · XS-101 — Agitator M-101 stopped with feed → unit trip

Requirement	Specification
Hazard (HAZOP N2.4, LOPA L2)	Benzene accumulates in an unmixed R-101; when mixing or hydrogen returns the held-up benzene reacts at once, R-101 boils at its ceiling and a vapour surge over-pressures the overhead and the loop.
Demand sources	• Agitator motor M-101 stops (motor protection, power) — 0.1 /yr • Agitator drive fails while the motor runs (coupling, shaft) — 0.01 /yr: not seen by this SIF Low-demand mode (frequencies from the LOPA tab).
Safe state	Benzene feed isolated (XV-101) and hydrogen stopped before the hold-up passes the limit.
Sensor and voting	M-101 motor run-status contact from the MCC; 1oo1. It proves that the motor is energised, not that the impeller turns.
Set point	motor stopped with XV-101 open · pre-alarm JI-101 LO 80
Delay	10 s (the model acts on the next 0.25 s step)
SIL target and basis	SIL 1, PFD ≤ 0.02. LOPA L2 (severity 4, target 10^{-3} /yr). Causes credited to this function: Agitator motor M-101 stops. Without it they occur at 0.05 /yr, so PFD ≤ 10^{-3} / 0.05 = 0.02 → SIL 1, and stricter than the SIL 1 band limit 0.1. Other causes in L2 that this function is not credited for: Agitator drive fails while the motor runs (coupling, shaft) (the motor still runs: the trip cannot see a failed drive); Recycle compressor K-101 stops (not this cause); Hydrogen flow lost with K-101 running (FIC-103 loop, FV-103 closes) (not this cause).
PFDavg (computed)	0.0106 (RRF 95, SIL 1 by PFD) OK · proof-test interval T = 1 yr, β = 0
Final elements	• XV-101 benzene SIS valve closes (fail closed, SOV) • XV-105 make-up hydrogen SIS valve closes (fail closed, SOV) • K-101 stopped (MCC stop relay) • BPCS actions, not credited: FIC-101, FIC-103, PIC-105 to MAN 0 %; E-101 to COOL with FV-104 forced 100 %; R-102 jacket to COOL
Arming and bypass	Armed only while XV-101 is open (benzene permitted), so start-up and shutdown without feed do not trip. Bypass: common requirements, §4.
Reset	Latched in SIS-100; manual reset on the SIS panel, refused while a non-motor trip condition is still present or TI-100 is above 210 °C; after the reset hydrogen and benzene stay isolated until they are admitted again through their permissives. A motor trip does not block the reset.
Proof test	Every year: stop M-101 with XV-101 open on a test permit (or simulate the contact), trip after 10 s; valve stroke and K-101 stop.
Process safety time and response	Governing demand: Agitator motor M-101 stops with benzene feed on. PST 1 min 31 s (trip point 0.25 s → hazard 1 min 31 s); response 12.75 s ≤ ½ PST. Hazard criterion: Benzene held up in R-101 above 11.8 kmol (its heat of reaction would take R-101 from 200 °C to its boiling point 258 °C; LOPA tab). Time verdict OK.
Finding	Blind to a drive failure with the motor running: the hold-up passes the limit with no trip (LOPA L2 cause 2; HAZOP dynamic check). Recommendation SL-R4 / R04: an agitator-failure trip on shaft speed SSL-101 or low power JT-101, 1oo2 (SIF-4R), replacing XS-101. Overall verdict Gap.

Reliability block diagram (LOPA tab, key xs101) — Sees the motor, not the impeller.

Subsystem	Architecture	λDU /yr	PFD	Share of PFDavg
M-101 run-status contact	1oo1	0.02	0.01	95 %
SIS-100 logic solver	1oo1	10^{-3}	5×10^{-4}	5 %
XV-101 + SOV or K-101 stop	1oo2	0.024 / 0.01	8×10^{-5}	1 %
SIF			0.0106	T = 1 yr

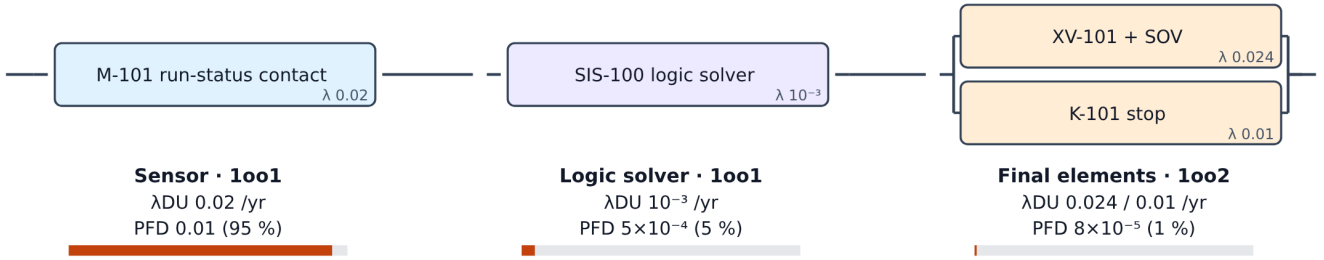


Figure 12 — SIF-4 XS-101 reliability block diagram (PFDavg = Σ subsystem PFD).

Process safety time on the model — 2 demand case(s); hazard criterion: Benzene held up in R-101 above 11.8 kmol (its heat of reaction would take R-101 from 200 °C to its boiling point 258 °C; LOPA tab).

Demand (applied at t = 0)	Trip point crossed	Hazard (no trip)	PST	Response	With the SIS acting	Verdict
Agitator motor M-101 stops with benzene feed on [governing]	0.25 s	1 min 31 s hold-up above 11.8 kmol	1 min 31 s	12.75 s	trip 10.25 s (XS-101); hold-up max 4.4 kmol; hazard not reached	OK response ≤ ½ PST
Mixing lost while the agitator motor keeps running (failed drive) [not seen]	—	1 min 31 s hold-up above 11.8 kmol	—	12.75 s	no trip; hold-up max 168.7 kmol; hazard reached 1 min 31 s	Gap the SIF does not see this demand: its trip point is never crossed before the hazard; the model still reaches the hazard with the SIS acting

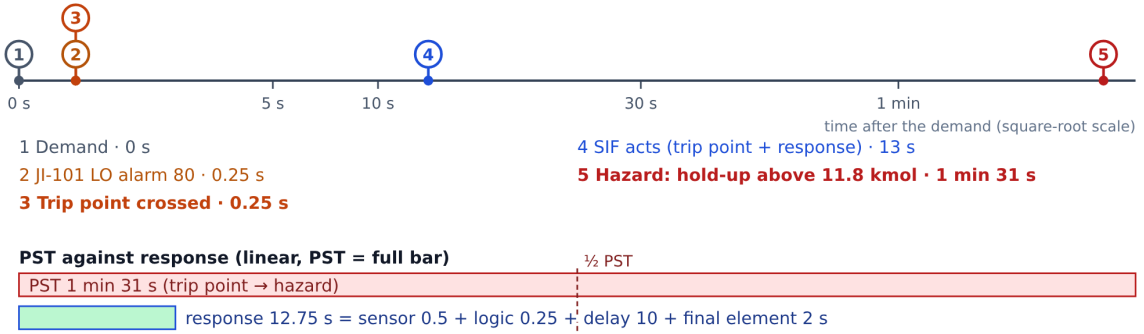


Figure 13 — XS-101 timeline: Agitator motor M-101 stops with benzene feed on (governing).

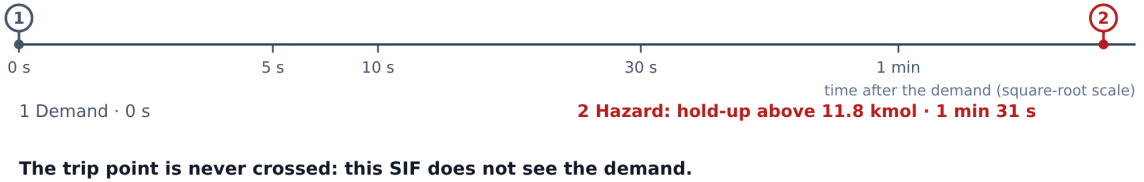


Figure 14 — XS-101 timeline: Mixing lost while the agitator motor keeps running (failed drive) (not seen by this SIF).

3.5 SIF-5 · XS-102 — K-101 stopped with feed → benzene isolated

Requirement	Specification
Hazard (HAZOP N8.2, LOPA L2)	Benzene accumulates in R-101 without hydrogen; when hydrogen returns the held-up benzene reacts at once (as SIF-4).
Demand sources	• Recycle compressor K-101 stops (motor, power, LSHH-104 action) — 0.1 /yr • Hydrogen flow to the sparger lost with K-101 running (FIC-103 loop, FV-103 closes) — 0.1 /yr: not seen by this SIF Low-demand mode (frequencies from the LOPA tab).
Safe state	Benzene feed isolated by XV-101 (K-101 is already stopped).
Sensor and voting	K-101 motor run-status contact from the MCC; 1oo1.
Set point	motor stopped with XV-101 open · pre-alarm FIC-103 LO 35000
Delay	10 s (the model acts on the next 0.25 s step)
SIL target and basis	SIL 1, PFD ≤ 0.02. LOPA L2 (severity 4, target 10^{-3} /yr). Causes credited to this function: Recycle compressor K-101 stops. Without it they occur at 0.05 /yr, so PFD ≤ 10^{-3} / 0.05 = 0.02 → SIL 1, and stricter than the SIL 1 band limit 0.1. Other causes in L2 that this function is not credited for: Agitator motor M-101 stops (not this cause); Agitator drive fails while the motor runs (coupling, shaft) (not this cause); Hydrogen flow lost with K-101 running (FIC-103 loop, FV-103 closes) (K-101 still runs: the trip cannot see the lost flow).
PFDavg (computed)	0.0225 (RRF 44, SIL 1 by PFD) Gap · proof-test interval T = 1 yr, β = 0
Final elements	• XV-101 benzene SIS valve closes (fail closed, SOV): the only acting element • XV-105 closes; K-101 already stopped • BPCS actions, not credited: FIC-101, FIC-103, PIC-105 to MAN 0 %; E-101 to COOL with FV-104 forced 100 %; R-102 jacket to COOL
Arming and bypass	Armed only while XV-101 is open. Bypass: common requirements, §4.
Reset	Latched in SIS-100; manual reset on the SIS panel, refused while a non-motor trip condition is still present or TI-100 is above 210 °C; after the reset hydrogen and benzene stay isolated until they are admitted again through their permissives. A motor trip does not block the reset.
Proof test	Every year: stop K-101 with XV-101 open on a test permit (or simulate the contact), trip after 10 s; XV-101 stroke time.
Process safety time and response	Governing demand: Recycle compressor K-101 stops with benzene feed on. PST 1 min 03 s (trip point 0.25 s → hazard 1 min 03 s); response 12.75 s ≤ ½ PST. Hazard criterion: Benzene held up in R-101 above 11.8 kmol (its heat of reaction would take R-101 from 200 °C to its boiling point 258 °C; LOPA tab). Time verdict OK.
Finding	Sees a stopped compressor only: a loss of hydrogen flow with K-101 running reaches the hold-up limit within minutes with no trip (LOPA L2 cause 4). Its single final element XV-101 also leaves the PFD above the LOPA number. Recommendation R24 with SL-R7: FSLL-103 on its own flow element, closing XV-101 and stopping the feed pumps (SIF-8R), replacing XS-102. Overall verdict Gap.

Reliability block diagram (LOPA tab, key xs102) — K-101 is already stopped, so XV-101 alone acts.

Subsystem	Architecture	λDU /yr	PFD	Share of PFDavg
K-101 run-status contact	1oo1	0.02	0.01	44 %
SIS-100 logic solver	1oo1	10^{-3}	5×10^{-4}	2 %
XV-101 + SOV	1oo1	0.024	0.012	53 %
SIF			0.0225	T = 1 yr

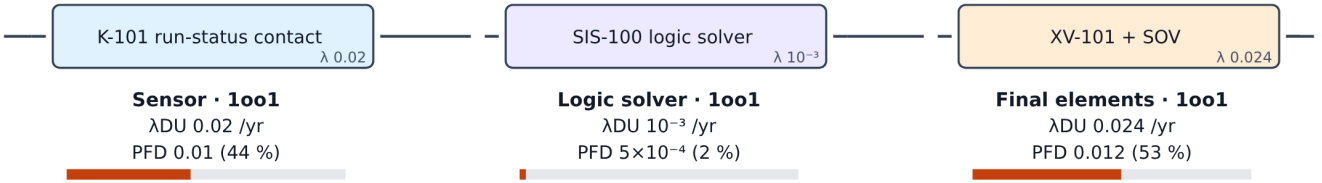


Figure 15 — SIF-5 XS-102 reliability block diagram (PFDavg = Σ subsystem PFD).

Process safety time on the model — 2 demand case(s); hazard criterion: Benzene held up in R-101 above 11.8 kmol (its heat of reaction would take R-101 from 200 °C to its boiling point 258 °C; LOPA tab).

Demand (applied at t = 0)	Trip point crossed	Hazard (no trip)	PST	Response	With the SIS acting	Verdict
Recycle compressor K-101 stops with benzene feed on [governing]	0.25 s	1 min 03 s hold-up above 11.8 kmol	1 min 03 s	12.75 s	trip 10.25 s (XS-102); hold-up max 5.4 kmol; hazard not reached	OK response ≤ ½ PST
Hydrogen flow to the sparger lost with K-101 running (FV-103 closed: FIC-103 output 0 %) [not seen]	—	1 min 06 s hold-up above 11.8 kmol	—	12.75 s	no trip; hold-up max 166.2 kmol; hazard reached 1 min 06 s	Gap the SIF does not see this demand: its trip point is never crossed before the hazard; the model still reaches the hazard with the SIS acting

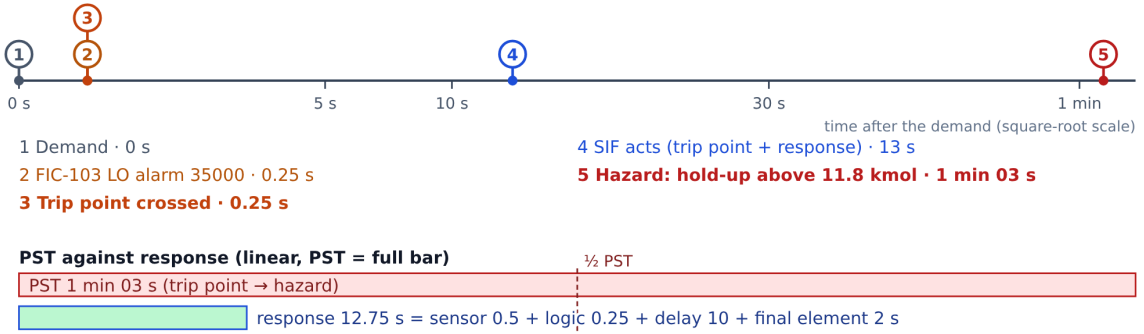


Figure 16 — XS-102 timeline: Recycle compressor K-101 stops with benzene feed on (governing).



Figure 17 — XS-102 timeline: Hydrogen flow to the sparger lost with K-101 running (FV-103 closed: FIC-103 output 0 %) (not seen by this SIF).

3.6 SIF-6 · LSHH-104 — V-104 level high-high → K-101 stops

Requirement	Specification
Hazard (HAZOP N7.4, N7.5)	Liquid carried from a full V-104 (100 %, assumed) into the running recycle compressor K-101: compressor damage, seal failure, hydrogen release.
Demand sources	• Product valve LV-104 closed (fails closed on air loss; LIC-104 transmitter reads low; LIC-104 in MAN) • Product line or tank inlet blocked Low-demand mode (frequencies from the LOPA tab).
Safe state	K-101 stopped before liquid reaches its suction; with feed on, XS-102 then trips the unit.
Sensor and voting	LT-104S, own level element (LIC-104 controls on LT-104); 1oo1
Set point	> 90 % · pre-alarm LIC-104 HI 78 (on LT-104, the control transmitter)
Delay	5 s (the model acts on the next 0.25 s step)
SIL target and basis	SIL 1, PFD ≤ 0.1. Not in the LOPA; SIL 1 assumed (typical practice for compressor protection with a release consequence; HAZOP N7.5, S4).
PFDavg (computed)	0.0105 (RRF 95, SIL 1 by PFD) OK · proof-test interval T = 1 yr, β = 0
Final elements	• K-101 stopped (MCC stop relay): the only action • With XV-101 open, XS-102 trips the unit 10 s later and becomes the first-out
Arming and bypass	Always armed. Bypass: common requirements, §4.
Reset	Not latched and no first-out: K-101 stays stopped; its restart is blocked while LT-104S is above 85 % (and while PIC-105 is below 10 bar g).
Proof test	Every year: raise the level on LT-104S (or simulate it), K-101 stops after 5 s; check the restart block above 85 %.
Process safety time and response	Governing demand: Product valve LV-104 closed (LIC-104 output 0 %, the fail position of LV-104). PST 3 min 34 s (trip point 14 min 19 s → hazard 17 min 53 s); response 7.75 s ≤ ½ PST. Hazard criterion: V-104 full (100 %, assumed) with K-101 running: liquid carried into the compressor. Time verdict OK.
Finding	Not taken to the LOPA: the SIL 1 target is assumed (machinery protection with a hydrogen-release consequence, HAZOP N7.5 severity 4). Take it to LOPA to confirm. Overall verdict OK.

Reliability block diagram (LOPA tab, key lshh104) — Not latched; with feed on, XS-102 trips the unit 10 s later.

Subsystem	Architecture	λDU /yr	PFD	Share of PFDavg
LT-104S (own element)	1oo1	0.01	5×10^{-3}	48 %
SIS-100 logic solver	1oo1	10^{-3}	5×10^{-4}	5 %
K-101 stop relay (MCC)	1oo1	0.01	5×10^{-3}	48 %
SIF			0.0105	T = 1 yr

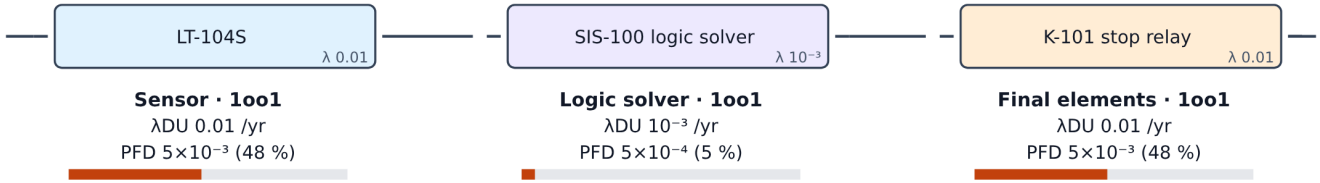


Figure 18 — SIF-6 LSHH-104 reliability block diagram (PFDavg = Σ subsystem PFD).

Process safety time on the model — 1 demand case(s); hazard criterion: V-104 full (100 %, assumed) with K-101 running: liquid carried into the compressor.

Demand (applied at t = 0)	Trip point crossed	Hazard (no trip)	PST	Response	With the SIS acting	Verdict
Product valve LV-104 closed (LIC-104 output 0 %, the fail position of LV-104) [governing]	14 min 19 s	17 min 53 s V-104 100 % with K-101 running	3 min 34 s	7.75 s	trip 14 min 35 s (XS-102); max 95.1 %; hazard not reached	OK response ≤ ½ PST

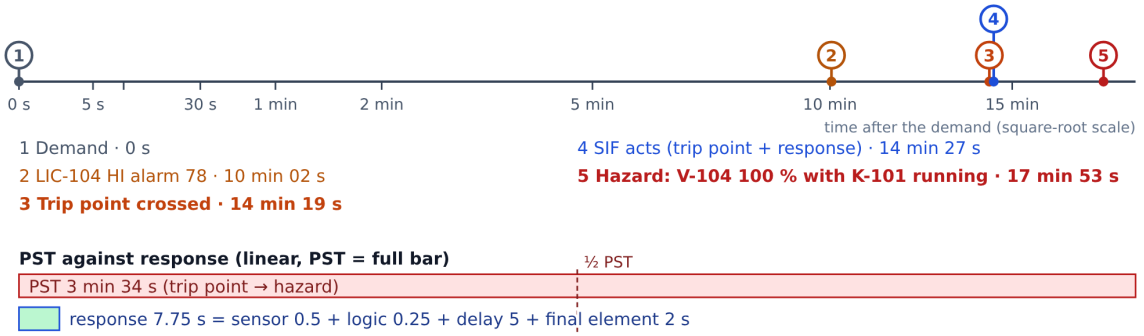


Figure 19 — LSHH-104 timeline: Product valve LV-104 closed (LIC-104 output 0 %, the fail position of LV-104) (governing).

3.7 SIF-7 · HS-100 — ESD pushbutton → unit trip and depressuring

Requirement	Specification
Hazard (HAZOP N9 and every node (manual ESD))	Any hazard the operator recognises first (fire, release, runaway): the unit must be isolated and depressured to the flare.
Demand sources	• Operator decision (fire, gas release, loss of control) Low-demand mode (frequencies from the LOPA tab).
Safe state	Unit tripped as for SIF-1 and EDV-101 open: the loop depressures to the flare.
Sensor and voting	HS-100 pushbutton on the console (hard-wired to SIS-100); 1oo1
Set point	pressed
Delay	none (manual)
SIL target and basis	SIL 1, PFD ≤ 0.1. Manual function, not credited as a SIF in the LOPA; designed to SIL 1 practice (assumed).
PFDavg (computed)	0.037 (RRF 27, SIL 1 by PFD) OK · proof-test interval T = 1 yr, β = 0
Final elements	• XV-101 benzene SIS valve closes (fail closed, SOV) • XV-105 make-up hydrogen SIS valve closes (fail closed, SOV) • K-101 stopped (MCC stop relay) • EDV-101 depressuring valve opens to the flare (only on ESD) • BPCS actions, not credited: FIC-101, FIC-103, PIC-105 to MAN 0 %; E-101 to COOL with FV-104 forced 100 %; R-102 jacket to COOL
Arming and bypass	Always armed. Bypass: common requirements, §4.
Reset	Latched in SIS-100; manual reset on the SIS panel, refused while a non-motor trip condition is still present or TI-100 is above 210 °C; after the reset hydrogen and benzene stay isolated until they are admitted again through their permissives. EDV-101 is closed again by hand on the SIS tab.
Proof test	Every year: press HS-100 on a shutdown; XV-101, XV-105 close, EDV-101 opens (stroke time), K-101 stops.
Process safety time and response	No process safety time: operator-initiated function.
Finding	Operator-started: it has no process safety time and in the LOPA it is part of the alarm-and-operator layer, not a separate SIF. All three valves must act (3oo3). Recommendation R29: depressuring study for EDV-101 and a fail-open actuator. Overall verdict OK.

Reliability block diagram (LOPA tab, key hs100) — Started by a person: in LOPA it is part of the alarm-and-operator layer, not a separate SIF.

Subsystem	Architecture	λDU /yr	PFD	Share of PFDavg
HS-100 pushbutton	1oo1	10^{-3}	5×10^{-4}	1 %
SIS-100 logic solver	1oo1	10^{-3}	5×10^{-4}	1 %
XV-101, XV-105 and EDV-101 (all must act)	3oo3	0.024 / 0.024 / 0.024	0.036	97 %
SIF			0.037	T = 1 yr

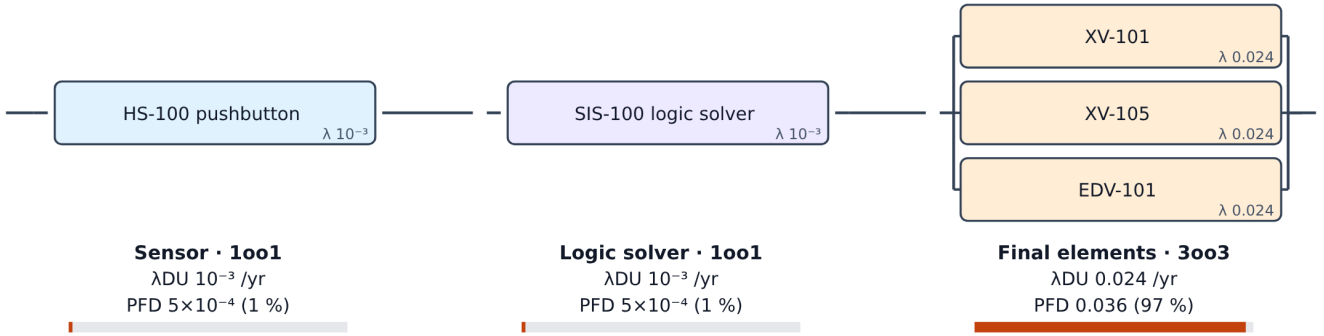


Figure 20 — SIF-7 HS-100 reliability block diagram (PFDavg = Σ subsystem PFD).

Process safety time: No process safety time: operator-initiated function.

3.8 PERM-H2 · AI-100 — Start permissive: XV-105 opens only with loop O2 below the limit

Requirement	Specification
Hazard (HAZOP N8.8, LOPA L4)	A hydrogen–air mixture inside the loop (oxygen above the limiting oxygen concentration, 5 % assumed) ignited by the pyrophoric Raney-Ni or static: internal explosion.
Demand sources	• Nitrogen purge incomplete before hydrogen is admitted — 0.02 /yr (operator error × 2 start-ups / yr, assumed) • Air drawn in after maintenance or nitrogen contaminated — 0.01 /yr (assumed) Low-demand mode (frequencies from the LOPA tab).
Safe state	XV-105 stays closed: no hydrogen enters a loop that still holds air.
Sensor and voting	AT-100 loop oxygen analyser (also the AI-100 HI alarm); 1oo1
Set point	≥ 1 % blocks XV-105 · pre-alarm AI-100 HI 1 (on the same analyser as the permissive)
Delay	none: checked when hydrogen is admitted
SIL target and basis	SIL 2, PFD ≤ 6.67×10^{-3}. LOPA L4 (severity 5, target 10^{-4} /yr). Causes credited to this function: Nitrogen purge incomplete before hydrogen is admitted (step skipped or not verified); Air drawn in after maintenance, or nitrogen contaminated with air. Without it they occur at 0.015 /yr, so PFD ≤ 10^{-4} / 0.015 = 6.67×10^{-3} → SIL 2, and stricter than the SIL 2 band limit 0.01.
PFDavg (computed)	0.0313 (RRF 32, SIL 1 by PFD) Gap · proof-test interval T = 0.5 yr, β = 0
Final elements	• XV-105 make-up hydrogen SIS valve stays closed (fail closed, SOV) • XV-101 stays closed too (benzene needs hydrogen admitted first)
Arming and bypass	Checked on every hydrogen admission (SIS command); not active once hydrogen is in the loop. Bypass: common requirements, §4.
Reset	Clears when AI-100 is below the limit; XV-105 is then opened by the operator command.
Proof test	Before each start-up (T = 0.5 yr, assumed): analyser calibrated with zero and span gas; with test gas above 1 % the open command for XV-105 must be refused; XV-105 seat-leak test.
Process safety time and response	No process safety time: a start permissive acts before the hazardous state exists.
Finding	One analyser carries the whole scenario: a stuck-low reading defeats both the permissive and its alarm (they share AT-100), and the PFD is above the LOPA number (L4, severity 5). Recommendation R26: a second analyser AT-100B (either one blocks), with LP-R1 XV-105B. Overall verdict Gap.

Reliability block diagram (LOPA tab, key o2perm) — Demanded at every hydrogen admission; analyser and valve proven before each start-up (T = 0.5 yr assumed, 2 start-ups / yr).

Subsystem	Architecture	λDU /yr	PFD	Share of PFDavg
AT-100 oxygen analyser	1oo1	0.1	0.025	80 %
SIS-100 logic solver	1oo1	10^{-3}	2.5×10^{-4}	1 %
XV-105 + SOV	1oo1	0.024	6×10^{-3}	19 %
SIF			0.0313	T = 0.5 yr

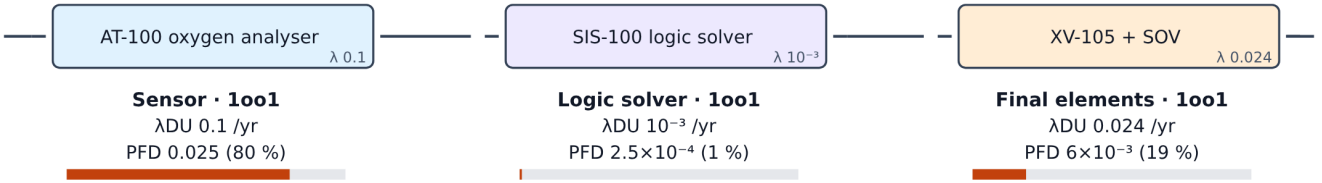


Figure 21 — PERM-H2 AI-100 reliability block diagram (PFDavg = Σ subsystem PFD).

Process safety time: No process safety time: a start permissive acts before the hazardous state exists.

4 · Common requirements

Requirements that apply to every function of SIS-100 (13 topics).

Topic	Requirement	Source
Mode of operation	Low demand: every demand source has a frequency of 0.1 /yr or less (LOPA) and the sum per SIF is below 1 /yr and below twice the proof-test frequency, so PFDavg is the measure.	IEC 61511-1 3.2.39, 10.3.2
Logic solver	SIS-100, separate from the DCS (own processor, I/O and power), SIL 3 capable (assumed); λDU 0.001 /yr. Trips latched in SIS-100 with first-out; the DCS only reads its status.	Safety layers tab; LOPA SIS-grade rates
Final elements	XV-101 and XV-105: tight-shut on/off valves, spring return, fail closed through a solenoid valve (SOV), not used for control; K-101 stopped by its MCC stop relay; EDV-101 fail open (assumed). FV-104, TV-102, HS-101 and HS-102 move on a trip but are BPCS valves and earn no SIL credit.	bmsTrip(); SL_FAIL
De-energise to trip	Every SIS input and output is normally energised; loss of power, air or signal trips the function (fail-safe). Exception: the ESD pushbutton is a normally closed contact.	Typical practice (assumed)
Response-time budget	Sensor 0.5 s + logic 0.25 s + trip delay + final element 2 s (assumed, to be confirmed by vendor data and stroke tests). Requirement: response ≤ ½ PST (OK); ≤ PST is Check; longer, Gap.	IEC 61511-1 10.3.2 (response time)
Proof test	Full function test from sensor to final element every year (T = 1 yr; the oxygen permissive before every start-up, T = 0.5 yr assumed), with stroke time and seat leakage of XV-101 and XV-105; proof-test coverage 100 % assumed; results recorded and failures fed back into the failure data.	IEC 61511-1 16.3; LP_SIF T
Bypass and override	Maintenance bypass by key switch on the SIS panel, one SIF at a time, alarmed and logged on the DCS, with a compensating measure (operator watching the pre-alarm) and a limit of one shift (assumed). XS-101 and XS-102 need no start-up bypass: they arm themselves when XV-101 opens.	IEC 61511-1 11.7.3
Reset	Manual and latched: the reset on the SIS panel is refused while a non-motor trip condition is present or TI-100 is above 210 °C. A reset does not restart anything: hydrogen is admitted again only with loop O2 below 1 %, benzene only with the eight feed permissives met.	bmsCmd('reset'); purgePermissives()
Operator interface	First-out on the SIS panel; SIS TRIP alarm and horn on the DCS; a pre-alarm below every trip setting (ALM); C&E matrix with a Live column; the trip timers on the SIS tab.	C&E matrix CHX-V1-IN-CE-001
Spurious trips	Target below 0.1 /yr per SIF (assumed). A 1oo1 sensor trips the unit on its own failure; 2oo3 voting (SL-R5) lowers both the PFD and the spurious-trip rate.	IEC 61511-1 10.3.2 (spurious trip rate)
Hardware fault tolerance	IEC 61511-1:2016 Table 6: SIL 1 and SIL 2 (low demand) need HFT 0, SIL 3 needs HFT 1. The PFD of SIF-2R and SIF-4R reaches the SIL 3 band only with β = 0; the SIL claimed here never exceeds 2.	IEC 61511-1 11.4
Failure data	SIS-grade dangerous-undetected rates of the LOPA tab (assumed typical values): transmitter 0.01 /yr, motor run contact 0.02, analyser 0.1, logic 0.001, XV with SOV 0.024, MCC relay 0.01, pushbutton 0.001; β = 0 (course method); PFDavg = λT/2 per 1oo1 item.	LOPA tab LP_SISR
Standards	IEC 61511-1:2016 clauses 10 (SRS), 11 (design), 16 (operation and maintenance); IEC 61508 for device certification; ISA-84.00.01; CCPS Layer of Protection Analysis.	

5 · Findings

5 of 8 functions have a Gap as found. The recommended functions of the LOPA tab close every PFD gap and see the demands the present trips miss.

Function	Verdict	What the check shows	Finding	Recommendation	With the recommendation
SIF-1 TSHH-100	Gap	• PFDavg 5.58×10⁻³ against ≤ 0.1 • PST 3 min 46 s against response 7.75 s • not seen: Reflux condenser E-103 without cooling water (HV-113 shut), pump-around running → hazard at 48 min 11 s with no trip of its own	Sees every loss of heat removal on the pump-around side, but not R-101 boiling down when the reflux condenser E-103 loses its cooling water: the temperature stays near 200 °C while the level falls (LOPA L1 cause 2). Recommendation R05 (LSLL-101 on its own element) or R06 (E-103 low cooling-water flow trip).	SIF-9R LSLL-101 (R05): R-101 level low-low on its own element → unit trip	PFDavg 5.58×10⁻³ OK against ≤ 0.1 Sees R-101 boiling down while its temperature is still normal.
SIF-2 TSHH-107	OK	• PFDavg 5.58×10⁻³ against ≤ 0.1 • PST 35 min 57 s against response 7.75 s	The trip elements are also the TI-107 indicators and the alarm, so a failed element blinds alarm and trip for its cell together. Recommendation SL-R5: separate SIS elements, 2oo3 per hot zone (TSHH-107S).	SIF-2R TSHH-107S (SL-R5): Separate SIS elements in R-103, 2oo3 per hot zone	PFDavg 6.8×10⁻⁴ OK against ≤ 0.1 Frees the TI-107 indicators and alarm from the trip.
SIF-3 PSHH-101	Gap	• PFDavg 0.0175 against ≤ 6.67×10⁻³ • PST 16.25 s against response 5.75 s	One valve XV-105 sets most of the PFD and the LOPA (L3, severity 5) asks for more than a 1oo1 valve gives. Recommendation LP-R1: a second make-up shut-off valve XV-105B in series (1oo2). Relief valves PSV-101 / PSV-104 (SL-R8, SL-R10, R02) remove the scenario almost entirely; a PI-101 pre-alarm (SL-R1, R21) helps the operator but shares PT-101.	SIF-3R PSHH-101 (LP-R1): PSHH-101 with a second make-up valve XV-105B (1oo2)	PFDavg 5.69×10⁻³ OK against ≤ 6.67×10⁻³ Two valves in series on the make-up: either one stops it.
SIF-4 XS-101	Gap	• PFDavg 0.0106 against ≤ 0.02 • PST 1 min 31 s against response 12.75 s • not seen: Mixing lost while the agitator motor keeps running → hazard at 1 min 31 s with no trip of its own	Blind to a drive failure with the motor running: the hold-up passes the limit with no trip (LOPA L2 cause 2; HAZOP dynamic check). Recommendation SL-R4 / R04: an agitator-failure trip on shaft speed SSL-101 or low power JT-101, 1oo2 (SIF-4R), replacing XS-101.	SIF-4R SSL-101 / JI-101 (SL-R4, R04): Agitator-failure trip: shaft speed or low power, 1oo2 → unit trip	PFDavg 7.13×10⁻⁴ OK against ≤ 0.02 Replaces XS-101: sees a stopped motor and a failed drive.
SIF-5 XS-102	Gap	• PFDavg 0.0225 against ≤ 0.02 • PST 1 min 03 s against response 12.75 s • not seen: Hydrogen flow to the sparger lost with K-101 running → hazard at 1 min 06 s with no trip of its own	Sees a stopped compressor only: a loss of hydrogen flow with K-101 running reaches the hold-up limit within minutes with no trip (LOPA L2 cause 4). Its single final element XV-101 also leaves the PFD above the LOPA number. Recommendation R24 with SL-R7: FSLL-103 on its own flow element, closing XV-101 and stopping the feed pumps (SIF-8R), replacing XS-102.	SIF-8R FSLL-103 (R24, SL-R7): Hydrogen flow low-low on its own element → XV-101 shuts, feed pumps stop	PFDavg 5.58×10⁻³ OK against ≤ 0.02 Replaces XS-102: sees any loss of hydrogen flow, motor running or not.
SIF-6 LSHH-104	OK	• PFDavg 0.0105 against ≤ 0.1 • PST 3 min 34 s against response 7.75 s	Not taken to the LOPA: the SIL 1 target is assumed (machinery protection with a hydrogen-release consequence, HAZOP N7.5 severity 4). Take it to LOPA to confirm.	—	—
SIF-7 HS-100	OK	• PFDavg 0.037 against ≤ 0.1	Operator-started: it has no process safety time and in the LOPA it is part of the alarm-and-operator layer, not a separate SIF. All three valves must act (3oo3). Recommendation R29: depressuring study for EDV-101 and a fail-open actuator.	—	—
Function	**Verdict**	**What the check shows**	**Finding**	**Recommendation**	**With the recommendation**
PERM-H2 AI-100	Gap	• PFDavg 0.0313 against ≤ 6.67×10⁻³	One analyser carries the whole scenario: a stuck-low reading defeats both the permissive and its alarm (they share AT-100), and the PFD is above the LOPA number (L4, severity 5). Recommendation R26: a second	PERM-H2R AI-100 / AI-100B (R26, LP-R1): Permissive with a second analyser (1oo2) and XV-105B	PFDavg 1.13×10⁻³ OK against ≤ 6.67×10⁻³ Either analyser above the limit blocks hydrogen; either valve isolates it.

Function	Verdict	What the check shows	Finding	Recommendation	With the recommendation
			analyser AT-100B (either one blocks), with LP-R1 XV-105B.		

5.1 Notes and open points

• Response times use assumed sensor, logic and valve times (0.5 + 0.25 + 2 s); confirm them with vendor data and stroke tests before the SRS is issued for design.
• The hazard limits are assumed where the design data do not give them (R-101 250 °C, R-103 260 °C, loop 36 bar g, V-104 100 %, benzene hold-up 11.8 kmol from the LOPA tab); a lower real design limit shortens every PST.
• β = 0 and no hardware-fault-tolerance check in the PFD numbers (course method); with β = 10 % the voted subsystems lose most of their advantage, and SIL 3 is never claimed.
• Demands a SIF cannot see (TSHH-100: E-103 without cooling water; XS-101: failed drive with the motor running; XS-102: hydrogen flow lost with K-101 running) are coverage gaps: the function exists, but the hazard arrives without its trip. They need the recommended functions, not a better PFD.
• Set points, delays and effects are identical to the C&E matrix CHX-V1-IN-CE-001; PSHH-101 (33.5 bar g) is below the relief set 36 bar g of the relief load basis CHX-V1-PE-RLB-001, which takes no credit for any trip.